SoK: Evaluating LLM and Agent Side-Channel Defenses From Evidence Audits to Real-Agent Measurements

***Abstract*—Encryption conceals content while exposing the structure of agent workflows. We systematize 82 study families through 39 source-bound result records, linking each defense outcome to its observer, execution boundary and evaluation conditions. This record-level view reveals three consequential transfer errors: attributing upstream protection to a plaintext-aware downstream emitter, treating discarded read cover as safe state-changing cover, and pricing one mechanism with another's utility or cost. The complete 58-core-family map and 20-family audit make these distinctions inspectable. Four experimental demonstrations connect the synthesis to deployment choices. Streaming reverses a buffered cost comparison; isolated and live cover have different state effects; real-agent traces separate behavioral compliance from operation privacy; and timing-aware comparisons expose when inexpensive transport shaping competes with costly shadow execution. The resulting contribution is a reusable evidence base and an operational method for selecting, comparing and validating defenses at the boundary where their protection is needed.**

1. Introduction

Encryption conceals message content, yet LLM engines and agents expose packet sizes, cache timing, visited domains and tool sequences [1], [2], [3], [4], [5], [6]. A defense result applies to a particular receiver, execution boundary and workload. Reusing a headline score across a frontend, dispatcher or tool provider can change both its protected signal and its cost object.

Our SoK uses the *evaluated result record* to make these transfers inspectable. Separating Dangol's stored-capture and live-protocol results prevents subtracting unmatched scores; separating Early Bird's KV-prefix and semantic-cache mechanisms prevents pricing one with the other's overhead. Combining signal origin with observer access and threat polarity explains why read-only privacy cover and state-changing bot evasion have different utility consequences. These findings emerge from binding evidence across records that a paper-level summary combines.

Contributions.

- **A populated evidence framework.** A complete map of 58 core and 24 adjacent families, 39 source-bound records and a nine-item audit of 20 mitigation families. Explicit rules identify compatible, incompatible and unresolved comparisons.
- **Three concrete transfer failures.** Frontend re-emission defeats an upstream-only protection claim; discarded retrieval cover does not justify live action cover; stored/live measurements and component costs cannot be interchanged. Source locators identify both the failed transfers and positive examples binding the required evidence.
- **Four demonstrations that change deployment choices.** Streaming/state tests and repeated agent trajectories from 53 customers expose adaptation rebounds, state side effects and timing leakage. Matched comparisons identify cases where inexpensive transport shaping competes with costly shadow execution, while the same audit separates measured logical conformance from end-to-end planner utility.

The synthesis connects established mechanisms—decoys, isolated execution, constant-rate tunnels and framing—to concrete evidence requirements for agent systems. Its distinctive output is a source-linked account of which transfers work, which fail, and which experiment resolves the difference (Table 1).

Scope of claims. We contribute record-level systematization and controlled demonstrations of evidence transfer. The real-agent target is a requested operation category; the known retail cohort contains 53 customers. Measurements establish fitted-suite inference, logical state conformance and scoped resource costs. Deployment-wide security and live defended-planner utility require the additional validation specified in §9.3.

Figure 1 places trace and action paths on the same agent–tool chain. This joint view makes the receiving representation and the execution boundary explicit before selecting a defense.

2. Scope and Systematization Method

2.1. Research questions, eligibility and threat polarity

We organize the systematization around four questions: which hidden targets, threat polarities and auxiliary observables appear across LLM and agent systems (RQ1); how signal generation origin relates to receiver observation permissions (RQ2); which holdouts establish portability across tasks, configurations and hardware (RQ3); and which evidence binds workflow protection, logical state preservation and resource cost, distinguishing these from full end-to-end planner utility (RQ4).

Core inclusion requires five conjunctive criteria: (C1) LLM or agent execution, (C2) a hidden target property,

TABLE 1. RESEARCH QUESTIONS, SYSTEMATIZATION FINDINGS AND EVIDENCE. EXPERIMENTAL FOLLOW-UPS TEST THE COMPARISON RULES; THE SOK CONTRIBUTION IS THE RECORD-LEVEL SYNTHESIS.

Question	Finding	Evidence in the main paper
RQ1: targets and observables	A common signal can serve confidentiality, attribution, evasion or authorized diagnostics; polarity changes the protected interest.	Complete core-family map (§4, Table 6).
RQ2: origin versus access	B-size cover does not bind A-call order; plaintext re-emission creates a fresh B trace; shared A-call projections do not establish unchanged state. These distinctions identify the evidence required to reuse a result.	NetEcho/SplitDecoy, WRA/AHB and dispatcher read/write records (§5); Exps 1–2.
RQ3: portability	A task holdout does not establish topology, hardware or model transfer; mixed configurations can preserve shortcuts.	Generator-specific holdouts (§7); customer-grouped analysis (§8.3).
RQ4: protection, state and cost	Bind inference, utility and cost to one scoped operating point. The 20-family audit gives R8 Bound 3/17 for component-or-task utility and 2/16 under a task-only sensitivity.	Twenty-family R1–R9 matrix; Exps 3–4 leakage/state/cost contrasts (§8). The same audit locates the utility boundary of our demonstrations.

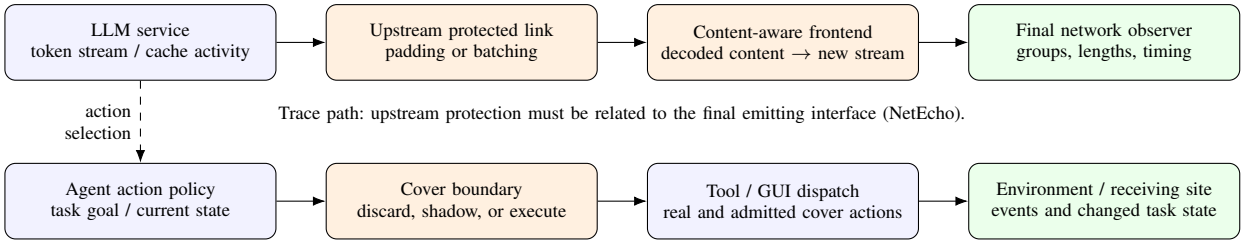


Figure 1. Observation and execution boundaries. The four demonstrations test emitter placement, stateful execution/link composition, real-agent operation signatures, and timing with matched transport controls.

(C3) an execution-dependent auxiliary observable rather than ordinary output semantics alone, (C4) specified observer permissions, and (C5) an empirical inference evaluation. Mixed-feature papers qualify only if they isolate an auxiliary-only C3 component; thus coding fingerprints, WildAuth, ZeroPrint, endpoint-content fingerprints and StepFinder remain adjacent boundary comparators [7], [8], [9], [10], [11]. Table 2 states the observable eligibility rules.

We use four narrative groupings of objectives and observer roles, not mutually exclusive classes. An unauthorized identity inference can be both attribution and a confidentiality concern. Each record separately specifies the inference target, observer, authorization relation and protected interest: **Type I** concerns confidentiality/privacy; **Type II** concerns model, harness or identity attribution; **Type III** concerns evasion of a platform's human–agent detector; **Type IV** concerns authorized operator diagnostics. Types can overlap. WRA protects research intent (I); UI Traces attributes agents (II); AHB evades a defensive detector (III); ClawGuard diagnoses loops (IV) [6], [5], [12], [13]. These roles are mapped per core family, alongside generating and receiving layers, in Table 6.

TABLE 2. OBSERVABLE-BASED ELIGIBILITY. A HIDDEN TARGET ALONE DOES NOT MAKE A SEMANTIC ATTACK A SIDE CHANNEL.

Observed input	Decision and required distinction
Length, timing, cache response	Core when measured execution dependence supports hidden-target inference.
Own-site actions	Core attribution with site permission; no automatic access to other sites or private tool logs.
Reply states the secret	Direct disclosure; adjacent, not auxiliary-channel evidence.
Reply style/content identifies a user	Adjacent unless a separately evaluated auxiliary mechanism is established. WildAuth stays a mixed-view comparator.
Output-filter membership oracle	Can be core: the filter's execution, not ordinary semantic output inference, mediates the signal.
Privileged loop/fault logs	Separate diagnostic component when C1–C5 hold; does not increase confidentiality evidence.

2.2. Corpus construction and reconciliation

We construct a purposive corpus through 40 seeds, venue-targeted proceedings screening and one-hop citation snowballing, supplemented by database and bulk keyword searches. The supplementary search returned 615 occur-

rences and 565 unique records; auditing 204 exclusions rerouted four records, yielding 15 new core and seven adjacent families. The final collection has 58 core and 24 adjacent families. The Boolean queries recover 25/60 previously indexed families (41.7% seed-set coverage); proceedings and citation routes supply complementary families. The assembled corpus and source versions define the scope of the synthesis. Appendix B provides the complete result-level index.

2.3. Coding procedure and consistency

AI assistance produced initial schema-guided field drafts. Authors checked and corrected all 39 result records and the 180 archived audit cells against source section/table/figure/paragraph locators, and reviewed all 82 family classifications using titles, abstracts and experimental sections. Every record carries a publicly inspectable source locator, enabling readers to check the exact result behind a comparison. The archive retains the initial extraction and author-adjudicated columns, disagreements and subsequent source corrections. Appendix A describes this audit trail.

SoK Takeaway 1 (Map evidence to the receiving deployment). A family-level label does not identify which signal, observer permission or protected objective a result supports. The complete core map separates signal origin from receiving access; the result ledger then binds each comparison to a source version and operating point.

3. Result Records, Comparability Rules and Validity

The atomic unit of our repository is one *evaluated result record*, rather than a whole paper or a single headline score. A result record binds a source version and locator to seven structured fields: Source, Target (and threat polarity Types I–IV), Receiver projection, Intervention boundary, Attacker training/calibration (D_0: clean-condition fit; D_2: protected-condition fit), Workload/holdout, and Outcomes (inference metric, task/component utility, and cost object; Table 3). A study family contributes multiple records whenever it evaluates distinct execution modes (e.g., stored capture vs. live loopback in Dangol [14]), distinct mechanisms (e.g., K-prefix KV simulation vs. GPTCache semantic de-identification in Early Bird [3]), or separately scoped performance benchmarks (e.g., SafeKV [15] and OTRO [16]).

Indexing schema and comparison contracts. The mnemonic SCOPE (Secret/target, Channel, Observer, Pipeline, Exploitation goal) names the target-and-channel columns of our index, paired with evaluation provenance. Generating layers span hardware (P1), serving (P2), transport (P3), application/API (P4), orchestration (P5), tools/MCP (P6), environment (P7) and multi-agent communication (P8). We apply three comparison contracts (Table 4): a comparison is *compatible* when required fields align; *incompatible* when a positive mechanism or boundary conflict is established (such as substituting token-payload ratios for wire bytes or offline timestamp edits for live action delays); and *unresolved* when a required binding is unreported (U), unavailable (A) or unverified (?). Table 5 illustrates a complete record-to-judgment lineage.

TABLE 3. A SINGLE RESULT RECORD. ONLY FIELDS RELEVANT TO THE REQUESTED COMPARISON MUST ALIGN; UNAVAILABLE REQUIRED FIELDS REMAIN UNRESOLVED.

Record field	Required meaning
Source	Family/result ID, version, source locator, reading scope and hash.
Target	Hidden property, inference goal and protection relation: confidentiality, attribution, human/agent distinction or authorized diagnosis.
Receiver	Actual permission and released representation, including session/document linkage and any auxiliary inputs.
Intervention	Where protection ends; whether edits affect stored traces, transmission or executed actions.
Attacker	Fit/calibration condition, test population, training access and adaptation budget.
Workload	Task, system, topology/hardware and grouping/holdout.
Outcomes	Inference metric / denominator; objective or proxy utility; applicable cost object; uncertainty and field states.

TABLE 4. COMPARISON RULES. A KNOWN MISMATCH AND A MISSING BINDING HAVE DIFFERENT CONSEQUENCES.

Requirement	Condition and resulting license
Descriptive comparison	Align target, receiver permission, metric meaning and cost object. Different systems may be described together; token payload and wire bytes cannot be ranked as the same cost.
Paired effect	Compare the same defended test units and outcome. A known test-unit mismatch makes that subtraction incompatible; an unverified fit binding leaves the effect unresolved.
Deployment transfer	Establish a bridge for the changed receiver, intervention, workload, topology or hardware. A demonstrated violation blocks the proposed transfer; an untested bridge leaves it unresolved. Permit only the range and endpoints actually covered.

Executable comparison rules. The record schema and comparison contracts precede the audit results. Table 8 gives all nine operational rules and a shared evidence-status alphabet; Table 9 applies them to every mitigation family. Twenty-four constructed cases and seven mechanism checks exercise rule application in the artifact, while the source-linked cases in §5 explain the substantive comparison judgments.

4. Corpus Map Across 82 Study Families

4.1. Origins, receiving projections and threat polarities

Across all 58 core families in our unified repository, 38 inherit runtime, serving or hardware channels (Origin B), while 20 concern agent-workflow channels arising from

TABLE 5. FILLED RECORD AND ITS LICENSED JUDGMENT. VALUES ARE SOURCE-REPORTED; NO CROSS-MODE EFFECT IS COMPUTED.

Join key / field	Entry
Family / result	Dangol workflow / `DANGOL-livebinding`.
Version / locator	arXiv:2606.07150v3, §9.7, live-protocol paragraph after Table 5 [14].
Target / view	Workflow classification from live Oblivious HTTP (OHTTP)/Agent-to-Agent (A2A) traffic; not GUI-state observation.
Intervention	Live fixed-rate cover with fresh per-session identifiers, distinct from transformed stored captures.
Workload / testbed	Workflow classes with client, relay, gateway and agent co-located on one loopback host.
Outcome / cost	Accuracy 0.30; about 13× bytes and 11× latency in that condition.
Unresolved fields	Objective terminal task utility U; precise paired live-refit comparison ?.
Licensed judgment	These costs describe this live protocol condition. They neither price dummy GUI actions nor prove task preservation. A matched objective endpoint is needed for the latter.

orchestration, tool dispatch, browser navigation and multi-agent protocols (Origin A; Table 6). Signal-generation origin is distinct from receiver location and threat polarity. The full map makes these distinctions visible family by family.

Cross-origin synthesis. Origin B includes token/packet lengths, cache timing and co-resident hardware channels; Origin A includes visited domains, tool dispatch and browser actions. Receiver permissions cut across these origins: a remote network observer, a same-site script and a privileged diagnostic hook expose different views even when they share an attribution target. Thus Origin A/B cannot substitute for Types I–IV or observer permissions. WRA's read-only privacy decoys, AHB's stateful evasion and SplitDecoy's exposed return path illustrate transfers that fail when these axes are collapsed [6], [12], [32]. The map retains every core family rather than using these examples as a substitute for coverage.

Layer incidence highlights where the systematization's comparison problem arises. In the 58-family map, signal-generation labels cluster at P2 serving/runtime (32 families) and P5 orchestration (19); receiving views cluster at P4 application/API (18), P1 hardware (16), and P3 transport (14). Generation at P3/P4 is rare (one family each), despite frequent observation at those layers. This source/receiver asymmetry makes the transfer question concrete: protection at a frequent source layer cannot be presumed to cover a different receiving projection. The map records layer sets, so it does not assign a causal edge to each source/receiver pair.

Purpose labels further show why a single "agent side channel" category is insufficient: Type I appears 41 times, Type II 13, Type III 2, and Type IV 5. The 32 map-only families extend coverage beyond mitigation comparisons: 15 list serving origins, 11 hardware and 11 orchestration; 19 concern confidentiality, nine attribution, one evasion and all five authorized-diagnostic families. They comprise 22 families without a scoped mitigation evaluation, five diagnostics, four unbound quantitative assertions, and SplitDecoy's boundary case (§6). Mapping these families locates privacy targets, operator capabilities and candidate validation environments even when no scoped mitigation record enters the ledger. Counts are overlapping family-label incidences; the audit then applies its reuse rules to the 39 scoped result records.

4.2. Relationship to prior systematizations

Prior surveys establish compound-system threat models and confidential-computing boundaries. Banerjee et al. categorize cross-layer compound AI threats under MITRE tactics; Forough et al. distinguish TEE isolation boundaries from side-channel resistance and note that inference protections need not cover agent memory or action coordination [64], [65]. Kotevska et al. propose a six-element privacy contract (asset, observer, channel, permitted disclosure, guarantee, evidence), with composition conditions framed as formal conjectures [66]. Classical website-fingerprinting studies establish bandwidth/latency costs for transport shaping, including Tamaraw, WTF-PAD, FRONT and Walkie-Talkie [67], [68], [69], [70]. Our contribution is to make reuse decisions inspectable across 82 mapped families and 39 source-bound result records, then organize the empirical checks around four boundaries: emitter and transport; stateful execution and the exposed link; repeated real-agent trajectories; and live timing with adapted website-fingerprinting controls. This evidence map shows why a defense's name or layer is insufficient for transfer: the receiver projection, state effects, workload and cost object must also align. Our experiments test these specific comparison obligations on a loopback TLS path and a pinned stateful tool simulator. These measurements connect established mechanisms to the cost and residual inference observed in streamed LLM and agent traffic.

5. Three Literature Case Studies

Record-level synthesis resolves three deployment decisions: whether upstream shaping survives another workflow hop, whether cover preserves the relevant state, and whether adaptive evaluation and cost describe the same mechanism. The following cases provide the source-local evidence for each judgment.

5.1. Case 1: bind the final target-bearing emitter

The downstream re-emission boundary. NetEcho's upstream packet-size shaping is an Origin B result; it does not bind the Origin A tool-call sequence. Its plaintext-aware frontend then emits a new Origin B trace (grouped token counts or character lengths), reopening a size channel after upstream protection [36]. SplitDecoy is a separate return-path record: the cloud sees nonzero gradient row support on every tested frame across nine seeds [32, Sections 5.1–5.3]. The inference is a changed receiver boundary, not one transferable score across B and A.

TABLE 6. COMPLETE MAP OF THE 58 CORE STUDY FAMILIES. EACH FAMILY APPEARS ONCE; LAYER AND TYPE LABELS MAY BE MULTIPLE AND DO NOT IMPLY ONE-TO-ONE CAUSAL PATHS. THE BLOCKS GROUP SOURCE LAYERS, NOT EVIDENTIAL QUALITY.

Core family	$G \to O$	Type
Origin B: hardware generation, P1 (8)		
Edge EM leakage [17]	$1 \to 1$	I
Energon [18]	$1 \to 1$	II
MoEcho [19]	$1 \to 1$	I
I Know What You Said [20]	$1 \to 1$	I
Behind Bars [21]	$1 \to 1$	I
PAGE404 [22]	$1 \to 1$	I
Spill the Beans [23]	$1 \to 1$	I
ArchSnoop [24]	$1 \to 1$	II
Origin B: hardware and service, P1/P2 (8)		
SparLeak [25]	$1,2 \to 1$	I
Shape and Substance [26]	$1,2 \to 1$	I
TDXRay [27]	$1,2 \to 1$	I
SparSEEty [28]	$1,2 \to 1$	I
OTRO [16]	$1,2 \to 1$	I
Detokenization Leaks [29]	$1,2 \to 1$	I
LLM-WFIN [30]	$1,2 \to 1,3$	I
LLM-serving RCA [31]	$1,2 \to 2$	IV
Origin B: service generation, P2 (13 of 22)		
SplitDecoy [32]	$2 \to 2$	I
KVGov [33]	$2 \to 2,4$	I
LLMs Have Rhythm [34]	$2 \to 3$	II
Remote timing [35]	$2 \to 3$	I
NetEcho [36]	$2 \to 3$	I
Prompt Echo [37]	$2 \to 3$	I
Speculative decoding [38]	$2 \to 3$	I
Remote keylogging [1]	$2 \to 3$	I
Whisper Leak [39]	$2 \to 3$	I
Cache contention [40]	$2 \to 4$	I
Routing metadata [41]	$2 \to 4$	I
SafeKV [15]	$2 \to 4$	I
ML privacy channels [42]	$2 \to 4$	I
Origin B continued: service generation, P2 (9)		
Early Bird [3]	$2 \to 4$	I
Prompt-cache audit [43]	$2 \to 4$	I
InputSnatch [44]	$2 \to 4$	I
Leaky Language Models [45]	$2 \to 4$	I
PromptPeek [2]	$2 \to 4$	I
SpliceLeak [46]	$2 \to 4$	I
Time Will Tell [47]	$2 \to 4$	I
CachePrune [48]	$2 \to 4$	I
MoE prompt stealing [49]	$2 \to 4$	I
Origin A: cross-layer sources (5)		
ClawGuard [13]	$1,5 \to 1$	IV
MIRAGE [50]	$2,5 \to 2$	IV
Cloud traffic benchmark [51]	$2,5,8 \to 3$	I,II
Web-agent unmasking [52]	$3,7 \to 3,7$	II,III
LumiMAS [53]	$4,5 \to 4,6$	IV
Origin A: orchestration/workflow sources (15)		
AgentPrint [4]	$5 \to 3$	I,II
Cycle detection [54]	$5 \to 4,6$	IV
SIGLEAK [55]	$5,6 \to 4,6$	I
ProcGrep [56]	$5,6 \to 4,6$	II
AgentProv [57]	$5,6 \to 6$	II
Lidar [58]	$5,6 \to 6$	II
Dangol workflow [14]	$5,6,8 \to 3,6,8$	I
Campaign attribution [59]	$5,6,8 \to 6,8$	II
WRA [6]	$5,7 \to 3$	I
PersonaFingerprint [60]	$5,7 \to 3$	I
Whose Agent? [61]	$5,7 \to 7$	II
AgentTell [62]	$5,7 \to 7$	I
UI Traces [5]	$5,7 \to 7$	II
FP-Agent [63]	$5,7 \to 7$	II
AHB [12]	$5,7 \to 7$	III

G denotes signal-generation layers and O the receiving observation layers. Layer indices: 1 hardware; 2 serving/runtime; 3 transport; 4 application/API; 5 orchestration; 6 tools; 7 browser/environment; 8 multi-agent communication. Types: I confidentiality/privacy; II attribution; III human–agent distinction/evasion; IV authorized diagnostics. These are objectives and observer roles, not mutually exclusive security guarantees. Privileged operator receivers and physical receivers are marked by the layer they inspect. Source versions, locators and normalization of prose-only layer descriptions are in the artifact.

The standard data processing inequality applies when $S \to Z \to Y$ is a Markov chain: target-oblivious post-processing cannot increase $I(S;Z)$ [71, Theorem 2.8.1]. A plaintext-aware frontend or unprotected backward computation changes that premise. The systematization insight is identifying which published records cross this boundary.

5.2. Case 2: resolve the WRA versus AHB cover asymmetry

Two axes of asymmetry: threat polarity and action risk. A comparison that groups WRA's decoy requests [6] and AHB's humanized actions [12] together as "cover traffic" can create an apparent paradox: WRA reports a clean privacy improvement (one concurrent decoy reduces functional-intent reconstruction E_{func} from 0.770 to 0.610), whereas AHB reports severe task collapse (Trip Planning success falls from 0.75 Raw to 0.15 under combined `Online+Fake` humanization).

Both records originate in Origin A actions, but polarity and state effects differ: WRA's concurrent decoy retrievals are discarded reads for Type I confidentiality, while AHB's live GUI events evade a Type III platform detector and can navigate or submit forms. Their utility outcomes are not contradictory, and AHB's missing condition-specific denominators preclude a matched causal effect. Origin A alone predicts neither receiver permissions nor whether an action mutates protected state.

Terminal-state reward does not establish dispatch-time authorization: an unconfirmed write could happen to produce the task's intended final state. For stateful tools, audit the real execution path and protected account/database, not only the terminal hash. Exp 2 (§8.2) tests this distinction with simulator mutations.

5.3. Case 3: bind adaptive retraining and cost objects

The $D_0 \rightarrow D_2$ rebound and component-cost boundaries. A matched adaptation contrast compares an undefended-data-trained attacker D_0 and a defended-data-trained, defense-aware attacker D_2 on identical defended tests. UI Traces' timestamp-delay refitting and Dangol's stored/live distinction show why receiver and run provenance must accompany the score [5], [14]. Exps 1–2 supply direct contrasts where D_0 reaches chance and D_2 recovers the signal; Exp 4 expands the attacker families.

On the cost side, Early Bird's 6ms/4% GPTCache overhead does not price K-prefix isolation; KVGov's $1.3\,\mu$s HMAC microcost omits lost cache reuse; and SpliceDefense's QCP cost does not price combined CTBF+QCP [3], [33], [46]. CachePrune binds attack suppression, QA/summarization utility and throughput; OTRO reports access-count leakage conditional on byte length and separately scoped TTFT/padding cost [48], [16]. Even within Origin A, identical call order is not a state guarantee: read/write traces can retain the same dispatch features while writes alter the account or database. The state claim therefore requires execution-path and final-state checks (Exp 2), not the Origin label or trace score alone.

6. Comparability Audit Across All 58 Core Families

6.1. Unified 58-core disposition and 39-record ledger

The corpus partitions as $82 = 58$ core $+24$ adjacent families. Within the core, $58 = 20$ mitigation $+6$ contextual-record $+32$ map-only families; the result ledger contains $39 = 33 + 6$ records, respectively, from the first two groups. The six are SIGLEAK, AgentTell, FP-Agent, Behind Bars, LLMs Have Rhythm and ML privacy channels [55], [62], [63], [21], [34], [42]. They supply fixed-guardrail, access-control, interference, VPN or filter-origin contrasts rather than the audited mitigation claim. The remaining 32 comprise 22 without a scoped mitigation evaluation, five diagnostics, four unbound assertions and SplitDecoy's boundary case. Appendix B lists all 39 records with source locators; Table 6 covers all 58 families. Table 7 groups mechanisms; Table 9 reports the 20-family audit.

6.2. Within-family and within-cohort record comparisons

While the machine-readable audit retains all $\binom{33}{2} = 528$ baseline pairs for completeness, 426 pairs cross unrelated mechanism cohorts and are outside scope by construction. The 13 within-family pairs are included in the 102 within-cohort pairs: $528 = 426 + 102$, with no additional subtraction of 13. Focusing on the **13 within-family pairs** (multiple records inside the same paper) and **102 within-cohort related pairs** (66 traffic-envelope, 28 cache-service, 3 behavioral-humanization, 3 generator-fingerprint, 1 local-memory, 1 fixed-instruction) reveals why paper-level comparisons fail:

- Even within the **13 same-family pairs** (plus 2 same-family pairs in the additions: `CACHEPRUNE-selective` vs. `CACHEPRUNE-context` and `OTRO-unpadded` vs. `OTRO-epochpad`), nearly half (6/13 baseline; 7/15 full) measure different primitives, targets or cost objects (e.g., Early Bird's K-prefix oracle simulation vs. its GPTCache semantic prototype; SpliceDefense's CTBF signal suppression vs. QCP-only prefill cost; OTRO's unpadded TTFT vs. derived epoch-padding cost).
- Among the seven aligned-outcome baseline same-family pairs, four evaluate different collection or execution populations (e.g., Dangol's stored-capture feature ablation at 0.17 accuracy vs. live loopback OHTTP at 0.30 accuracy) and three lack a verified matched-test binding, so none licenses cross-record subtraction as a single paired deployment effect.
- Across the **102 related-cohort pairs**, only 16 (15.7%) support a source-qualified descriptive comparison, whereas 80 (78.4%) exhibit positively established receiver, target or cost-object mismatches and 6 (5.9%) remain unresolved.

6.3. Nine-item split reporting audit across 20 mitigation families

Table 8 defines each item before its application to the 180 cells in Table 9. Applicability follows the result's reuse claim: a fixed analytical attack can make adaptation inapplicable, while a component benchmark can support component utility. U and ? retain distinct source states. The 11 unresolved adaptation cells comprise five R4 and six R5 bindings, explicitly distinguished from an attack failure.

For Dangol, R5 is Bound for in-fold stored-capture experiments; the live-refit paired comparison in Table 5 remains ?. These are different records, so we retain the family count of two and explicitly restrict its meaning.

SoK Takeaway 2 (Evidence needed for reuse). The audit distinguishes component-or-task utility (Bound 3/17 applicable families) from deployment-task utility (2/16 when TDXRay's lookup microbenchmark is treated as inapplicable); these are scope sensitivities, not quality ranks. R5 binds matched adaptation for 2/11 applicable families; a narrower learned-adaptive-attack denominator excluding Shape and Substance is 2/10. CachePrune binds inference, utility and cost together; it does not thereby establish adaptation. The record index in Appendix B identifies the underlying operating points.

7. Match Transfer Claims to the Generator

7.1. Different tests rule out different explanations

A fingerprint is generated by a model interacting with tasks, prompts, tools, browser state and serving infrastruc-

TABLE 7. EIGHT INTERVENTION FAMILIES ORGANIZED BY THREAT POLARITY (TYPE I: CONFIDENTIALITY PROTECTION; TYPE II: PROVENANCE/IDENTITY ATTRIBUTION; TYPE III: BOT-DETECTION EVASION), RECEIVING VIEW AND SUPPORTED DEPLOYMENT JUDGMENT. BECAUSE OBJECTIVES AND RISK MAGNITUDES DIFFER, THESE ROWS ARE NOT A COMMON EFFICACY RANKING.

Intervention & polarity	Evaluated view, target and source	Established result and applicable boundary
Padding / batching (Type I: confidentiality)	Network packet lengths and groups; topic inference. NetEcho and Whisper Leak [36], [39].	Collected padding traces retain inference signal; packet-group batching is evaluated by replay. A protected upstream length view does not cover downstream re-emission or residual grouping. Live incremental cost remains U.
Constant-rate / transport cover (Type I: confidentiality)	Network workflow fingerprint; Dangol's OHTTP testbed [14].	The co-located loopback setting yields 0.30 recovery at about $13\times$ bytes and $11\times$ latency. Transformed-capture results are separate; neither result supplies objective agent-task utility.
Concurrent query decoys (Type I: confidentiality)	Destination/domain observer; private prompt and trait inference. WRA [6].	Real/decoy research prompts execute in parallel as read-only queries whose decoy answers are discarded locally; interleaved-domain evaluation reduces one-decoy E_{func} from 0.770 to 0.610. Terminal decoy utility/cost are U.
Action humanization (Type III: bot-evasion)	Mobile GUI touch traces; evading human–agent bot detection. AHB [12].	Unlike WRA's read-only privacy decoys, AHB's `Fake` mode dispatches live mobile GUI touches to evade platform bot detectors, dropping Trip Planning success from 0.75 to 0.15. Non-Fake `History+Long` (0.7143 success) avoids GUI mutation.
Cache isolation / keying (Type I: confidentiality)	Co-tenant prefix timing; SafeKV, KVGov and CachePrune [15], [33], [48].	SafeKV reports post-trigger AUC near 0.5 and separate TTFT/throughput comparisons. KVGov's HMAC median 1.3 μs prices key computation, excluding lost reuse. CachePrune jointly binds marked-token protection, QA/summarization utility and serving performance.
Cache timing equalization (Type I: confidentiality)	Co-tenant cache-fusion timing; SpliceDefense [46].	Constant-Time Boundary Fusion (CTBF) suppresses the tested signal over 15 tasks/75 steps. Quantized Chunk Padding (QCP) adds mean 7.5 tokens and < 5ms/$< 1\%$ prefill time; this does not price combined deployment.
Timestamp perturbation (Type II: attribution)	Recorded website action metadata; model/agent attribution. UI Traces [5].	Fitting on delayed traces (D_2) recovers substantial attribution performance under offline timestamp edits that retain action types/order. Live execution delay and task outcomes are not measured by the trace transformation.
Oblivious / unshared memory (Type I: confidentiality)	Host or local resource observations; TDXRay, Gao et al. and OTRO [27], [20], [16].	TDXRay measures oblivious-map lookup cost; Gao measures loading/memory cost after disabling zero-copy. OTRO reduces conditional access-count information (3.94 to 0.04 bits) with separate unpadded TTFT (4.59%) and epoch-padding (2.8ms) bounds.

Authorized operator diagnostics (Type IV: ClawGuard, cycle detection, LumiMAS, LLM-serving RCA and MIRAGE) are baseline telemetry rather than privacy attacks or evasion defenses, and are cataloged separately.

ture. A holdout establishes portability along the axis it changes: new tasks, browser configurations, network topologies and hardware are distinct deployment tests.

UI Traces fixes a browser/action harness while testing tasks and environments. FP-Agent combines behavior with static browser attributes; shared visitor proxies are not authenticated independent people or hosts [5], [63]. Task separation can remove memorization of those task instances while retaining configuration identifiers. This bounds portability to the tested configuration. The traffic benchmark separately tests generating topology, so its shift is a different test rather than a stronger score on the same estimand [51].

LeakyLMs illustrates the boundary for system-property rather than agent identity. Its deployment probe observes streaming events through an ordinary client API. Its architecture branch additionally assumes known GPU type, same-class calibration and single-GPU execution [45]. Remote architecture inference therefore does not establish hardware-agnostic inference. The joint search in its remote example ranks the true configuration 28/2068, distinct from a conditional parameter search or a near match; those outcomes must not be collapsed into universal exact recovery. Source-reported local architecture generalization and remote deployment signatures remain positive evidence under their own assumptions.

7.2. Representation increment is not portability

Within a fixed observer and test population, coarse counts versus sequence features test whether the latter improve a specified predictor. WRA already uses ordered domains before adding timing; Kang changes class sets across some conditions [6], [61]. These test different contrasts from an order-only effect. A representation comparison therefore holds target labels and evaluation units fixed while changing the observed features.

Adjacent coding-agent PR attribution depends on repositories, submission pathways and artifact generation. Its mixed-feature result does not isolate an auxiliary-only C3 view; inspected code computes weighted F1 and reduces features before cross-validation, but the exact paper-run binding remains unverified [7]. This supports an adjacent attribution comparison under its reported inputs. WildAuth is adjacent under the same rule; input/partition ambiguities do not supply a model-independent response-only privacy claim [8].

7.3. Resulting claim hierarchy

Task-specific identification, task-disjoint identification in a fixed harness, configuration/topology transfer and hardware-calibrated architecture inference require different evidence. Origin A/B labels locate generating layers; they do

TABLE 8. OPERATIONAL RULES FOR THE NINE REUSE BINDINGS. APPLY EACH ITEM TO A SPECIFIED RESULT RECORD AND DEPLOYMENT CLAIM.

Item	Evidence required for a Bound (B) judgment
R1 Target	Hidden property, inference goal and authorized/protected interest are explicit.
R2 Receiver	Observer permission, released representation and auxiliary inputs identify the evaluated view.
R3 Intervention	Placement and execution mode bind the transform to stored data, wire emission or executed actions.
R4 Protected fit	Attacker fitting/calibration uses the protected condition; training and protected testing are identified.
R5 Matched adaptation	Clean- and protected-condition fits are evaluated on the same defended test population.
R6 Workload	Tasks, system/generator configuration and holdout specify the supported transfer axis.
R7 Inference	Metric, denominator and evaluation condition bind the attack outcome to the scoped mechanism.
R8 Utility	Task/component performance is measured at the same protected operating point, with its state boundary explicit.
R9 Cost	Resource units, baseline and measurement object price that operating point.

B: bound; P: partial binding; U: unreported in inspected scope; A: source unavailable; ?: binding unverified; –: inapplicable. First determine applicability, then assign evidence status. Family B means at least one scoped record satisfies the item, not that every result does.

TABLE 9. PER-FAMILY EVIDENCE BINDINGS FOR ALL 20 EVALUATED-MITIGATION FAMILIES. R1–R9 ARE SCOPED REUSE ITEMS, NOT OVERALL PAPER SCORES.

Family	R1	R2	R3	R4	R5	R6	R7	R8	R9
UI Traces [5]	**B**	**P**	**B**	**B**	**B**	**B**	**B**	U	–
WRA [6]	**B**	**P**	**B**	–	–	**P**	**B**	**P**	U
NetEcho [36]	**B**	**B**	**B**	**B**	**P**	**B**	**B**	U	U
Early Bird [3]	**B**	**B**	**B**	**B**	**P**	**B**	**B**	U	**B**
Whisper Leak [39]	**B**	**B**	**P**	?	?	**B**	**B**	U	**B**
SpliceLeak [46]	**B**	**B**	**B**	–	–	**B**	**B**	U	**B**
Time Will Tell [47]	**B**	**B**	**B**	?	?	**B**	**B**	U	**B**
I Know What You Said [20]	**B**	**B**	**B**	–	–	**B**	U	–	**B**
Speculative decoding [38]	**B**	**B**	**P**	?	?	**B**	**B**	–	**B**
Dangol workflow [14]	**B**	**B**	**B**	**B**	**B**	**B**	**B**	U	**B**
Cloud traffic benchmark [51]	**B**	**P**	**B**	?	?	**P**	**B**	**P**	U
Remote timing [35]	**B**	**B**	**B**	–	–	**B**	**B**	U	**P**
TDXRay [27]	**B**	**B**	**B**	–	–	**B**	U	**B**	**B**
SafeKV [15]	**B**	**B**	**B**	–	–	**B**	**B**	U	**B**
AHB [12]	**B**	**B**	**B**	**P**	?	**B**	**B**	**B**	U
KVGov [33]	**B**	**B**	**B**	–	–	**B**	**B**	U	**B**
CachePrune [48]	**B**	**B**	**B**	–	–	**B**	**B**	**B**	**B**
OTRO [16]	**B**	**B**	**B**	–	–	**B**	**B**	–	**P**
SparLeak [25]	**B**	**B**	**B**	?	?	**B**	**B**	**P**	U
Shape and Substance [26]	**B**	**B**	**P**	U	U	**B**	**P**	U	**P**
Bound	20	17	17	4	2	18	17	3	11
Applicable	20	20	20	11	11	20	20	17	19

B: bound; **P**: partial; U: unreported in inspected scope; ?: unverified binding (11 cells); –: inapplicable. Table 8 defines R1–R9 and the shared status alphabet. A bound cell is supported by at least one scoped record. Dangol R5 concerns stored capture; its live-refit binding remains unverified. CachePrune jointly binds R7–R9 at one operating point. Source follow-up resolves cloud-traffic R3 to B and Shape and Substance R4/R5 to U.

not establish a receiver view or cross-layer equivalence. Our case synthesis uses them to correct three tempting transfers: upstream B-size cover does not bind A-call order, plaintext re-emission creates a fresh B trace, and an unchanged A-call projection does not prove state invariance. Together they identify which evidence must travel with a reuse claim. For portability claims, change the relevant generator while preserving target labels and receiver contract; for representation increments, compare identical units and disclose predictor/selection restrictions.

8. Four Experimental Demonstrations

Four experiments turn the synthesis into testable comparison obligations: **(1) emitter placement**, **(2) execution state and link composition**, **(3) real-agent operation signatures**, and **(4) timing and matched traffic controls**. Each holds a boundary fixed while changing the mechanism relevant to that obligation. Table 10 identifies the three real-agent estimands; Appendix C maps the modules to archived study identifiers and extensions.

Shared measurement conventions. A byte-transparent relay captures post-handshake TLS application records on loopback; TLS 1.3 session tickets are disabled. Standard-prompt results use the arm called `baseline` in the artifact; *undefended* denotes transport without a program defense. Balanced accuracy (BA) averages the two class recalls. We report the maximum over each specified attack suite and recompute that maximum in each of 2,000 common bootstrap draws, grouping tasks for reference-program tests and customers for real-agent tests. The original protocol additionally screens for an upper 95% BA bound below 0.60 and maximum AUROC below 0.65. We retain this engineering threshold, reporting its interpretation once in §9.3.

TABLE 10. COHORT, OBSERVATION AND COST DENOMINATORS FOR EXPS 3–4. STANDARD-PROMPT PAIRS: CANCEL/RETURN BA (%). COUNTS COVER BOTH PROMPTS; BYTE RATIOS PRICE JOINT CONSTANT SHAPING AGAINST EACH STUDY'S OWN UNDEFENDED SERIALIZATION.

Study	Observation / fit	Raw BA	Bytes×
Exp 3 frozen	Replay lengths/order; frozen suite	97.2/94.0	15.39
Exp 3 cross-fit	Analytic lengths/order; 11-model cross-fit	91.7/85.5	19.76
Exp 4	New RPC envelope; lengths/order/time	91.7/86.4	15.11

Frozen-split and RPC rows: 276 test episodes, 24 customers, 5/7 positive customers. Exp 3 cross-fit: 690 episodes, 53 customers, 14/18 positive customers. Each prompt has half the episodes. Exp 3 cross-fit fold calibration yields larger templates (5.04× calls); frozen-split and RPC rows use 3.83×. Repeats measure execution variation within these customer groups.

TABLE 11. EXP 1: THE SAME 48 SAVED RESPONSES EMITTED BY A BUFFERED OR A TOKEN-STREAMING FRONTEND (TLS 1.3 LOOPBACK CAPTURES). BYTES ARE RELATIVE TO THE UNPADDED EMITTER OF THE SAME KIND. TTFB/DONE: FIRST AND LAST DELIVERED BYTE IN TOKEN INTERVALS (ACCOUNTING PROXY). ACCURACIES ON 24 HELD-OUT OUTPUTS (12 TOPICS); ID: CLOSED-WORLD TOP-1 IDENTIFICATION AMONG 48 RESPONSES (CHANCE 2.1%).

				Verbosity		Prefix	
Emitter / framing	Bytes×	TTFB	Done	D_0	D_2	D_2	ID
Buffered, unpadded	1.00	105	105	100.0	100.0	91.7	87.5
Buffered, training cap	2.19	105	105	50.0	50.0	50.0	2.1
Stream, per token	1.00	1	105	100.0	100.0	87.5	100.0
+ random event pad	1.29	1	105	100.0	100.0	87.9	62.3
+ constant event size	1.23	1	105	100.0	100.0	87.5	58.3
8-token batches, const.	0.21	8	105	50.0	100.0	87.5	20.8
Const. events + count cap	2.42	1	208	50.0	50.0	50.0	6.3

D_2 is the best of total-byte threshold, record-count threshold and a logistic sequence model, each fitted on defended training outputs. Random padding: 10 socket-measured draws plus an independent profiling run. TLS 1.2 gives identical accuracies and byte ratios within 0.03. Verbosity contrasts brief/expanded outputs; Prefix contrasts the first two/three sentences of each expanded output. Tokens use o200k_base. Training count caps are 208/88 events, with 2/24 and 5/24 test overflows, respectively.

8.1. Exp 1: bind protection and cost to the emitter

Buffered tests use brief/expanded saved responses for 24 topics (12 training, 12 test). Padding only the upstream link leaves verbosity 100% separable at a plaintext-aware terminal emitter; padding both reduces BA to 50%, reproducing NetEcho's mechanism [36]. A training-fitted buffered cap achieves 50% at 2.18× wire bytes under TLS 1.2 (2.19× under TLS 1.3).

Streaming the same 48 texts as per-token server-sent events changes the cost contract [1], [39]. Per-event padding retains token count and 100% verbosity inference. Eight-token batching cuts bytes to 0.21× and clean-trained BA to 50%, yet defended-data refitting restores 100%. Constant shape reaches 50% at 2.42× bytes and 208 versus 105 token intervals; two overflowing test outputs remain identifiable (Table 11). **Finding.** The same saved content yields different protection and delay under buffering and streaming. A buffered cap's byte ratio cannot price a streaming deployment, whose first-output delay and emission schedule are different outcomes.

8.2. Exp 2: compose state preservation with link protection

Execution boundary. Among 115 public retail tasks, 97 have an initially pending order for the task user. Three order-selection seeds yield 291 cover placements before the reference program. Same-item modification changes status and appends a refund in 173 placements despite unchanged item and price; the other 118 calls fail the variant-availability check. Idempotence, $f(f(s,a),a) = f(s,a)$, therefore differs from state preservation, $f(s,a) = s$. Cancelling another user's order changes the shared database. The shadow executor isolates cover effects on a separate state and suppresses its responses; a live-execution control exposes the resulting side effects.

TABLE 12. EXP 2 MAIN DISPATCH RESULTS: ALL 115 REFERENCE PROGRAMS, THREE SEEDS, KNOWN-POPULATION DEFENSE CALIBRATION; TASK-FOLDED ATTACKERS. MAXIMUM BA (CANCEL/RETURN); EXTRA CALLS PER REAL CALL; DATABASE EQUALITY.

Policy	BA (%)	Calls+	DB eq.
No cover	100.0/100.0	0.00	100%
Cancel, $p = .5$	86.1/100.0	.10	100%
Cancel, $p = 1$, shadow	100.0/100.0	.20	100%
Cancel, $p = 1$, live	100.0/100.0	.20	0%
Cancel count → 2	76.9/100.0	.35	100%
One decoy program	91.9/85.9	1.01	100%
Two decoy programs	83.5/78.1	1.98	100%
Shape equalization	50.0/54.5	6.71	100%

The dispatch observer sees ordered tool names and response lengths. Policies add cancellation, pad its count, interleave donor programs or equalize all tool counts; donors/templates use the known 115-program population. Task-grouped five-fold attacks fit clean/defended gradient boosting and logistic regression on counts, bigrams and lengths, with a count-rule baseline and three seeds (18/32 positive cancellation/return tasks). One inserted cancellation drives clean-trained BA to chance, while refitting restores 100%. Count equalization removes that rule's signal, but correlated tools retain 76.9% BA. Donor programs, inspired by WRA's discarded retrieval cover [6], leave 83.5–91.9%. Shape equalization reaches 50.0/54.5% at 6.7 extra calls per real call (Table 12). Shadow checks match the original live state in every case; live cancellation changes it in every case.

Final network view. Four cover policies and three framing choices produce 4,140 TLS 1.2 sessions. The network observer receives direction-tagged record lengths and order. A training-fold profiler maps request sizes to tool types before applying dispatch attacks: request-only identification reaches 89.9/91.4% (582 calls; majority 29.4%). Per-call buckets leave return BA at 73.9%; joint shape cover and constant framing reduce cancellation/return to 54.4/50.0%, at 26.3× bytes and 7.7× calls.

Finding. Call-level cover and per-call framing address complementary observables. Full constant-rate transport can instead hide packet counts from a passive link observer without executing dummy tools [72]; Exp 4 measures that alternative. At a dispatch view after decapsulation, cover requires hidden markers and routing to the trusted isolated executor (§9.1). The intervention must match the receiver and the protected state.

8.3. Exp 3: real-agent signatures and calibrated shaping

Task and collection. Cancellation and return label the reference program's requested

operations, `cancel_pending_order` and `return_delivered_order_items`. This operational target tests whether encrypted metadata reveals which service action a task requests. GPT-6.1 Sol/low plans from public retail policies, schemas and actual sandbox observations; a separate context simulates the user. Two prompts and three repeats on 115 tasks yield 690 runs from 53 customers. The rules prompt adds a customer-details read before order access and a read before each write.

Frozen calibration and behavioral control. All 414 training runs (69 tasks) precede freezing shared cover templates, donors and framing caps. The 276 test runs cover 46 tasks from 24 disjoint customers. Prefix-only scheduling inserts training-derived cover, flushing unmet counts at episode end. Four program conditions per prompt give 2,760 local TLS replays with lengths/order attacks; every failure and handoff remains included. Real-call pass-through plus isolated cover preserves the logical response/state projection (Appendix A); reusing trajectories pairs the network interventions on identical model decisions.

Rules raise read-order adherence from 86.2% to 97.1%, while undefended category inference remains above 93%. Native rewards are 110/138 and 105/138 (paired difference -3.6 points, 95% interval $[-9.5, 1.4]$). Thus compliance and metadata privacy separate in the same task population. Joint cover and framing reach 50.0/50.0% point BA, compared with 97.2/94.0% undefended; their upper bounds are 60.6/65.5%. The paired decreases are 47.2/44.0 points (Table 13). The rules-arm joint return point is 60.6%, identifying prompt/cohort sensitivity.

Customer-grouped exploratory comparison. Five-fold cross-fitting refits templates, donors, caps and all attackers on other customers, pooling the full known cohort (14/18 positive customers). The analytic TLS model matches all 2,760 saved sessions and 120 additional captures. An 11-model union includes a 1D-CNN and k-FP-style forest-leaf learner. Joint constant shaping reduces BA by 31.2/30.0 points (paired 95% intervals [15.3,41.0]/[14.8,39.2]); buckets reduce it by 34.6/32.8 points, at lower byte cost (Table 14).

Locate the observable operation signature. An oracle selects write exchanges using true tool types, then exposes only ciphertext lengths/order. Writes comprise 19.6% of real calls yet retain 91.7/84.9% BA; removing them leaves 62.8/59.9% (Table 15). The target names an operation, so the result locates its metadata signature. It directs protection toward specific exchanges, while residual read inference shows why hiding one named call is insufficient. **Finding.** Better read-order compliance leaves operation inference intact; calibrated joint shaping reduces that fitted-suite inference; and the localization identifies which exchanges carry it. Semantic-attribute generalization is a separate estimand, tested by the small extension summarized in Appendix C.

8.4. Exp 4: timing, latency and matched transport controls

All 690 saved action trajectories execute through a live local tool server under six conditions, producing 4,140 new TLS sessions. Each real response gates the next action; real and shadow calls have separate states. The original 414/276 customer-disjoint split is retained, with attacks fitted and frozen before test capture. Lengths, directions and arrival times measure the tool RPC chain.

A complete tunnel sends 4,096-byte cells in both directions at 2 or 5 ms target intervals; dummy cells execute no tools. A training-sized 133-slot horizon continues after the last response. Serial and two-in-flight shadow schedules use the frozen prefix-only count policy with constant per-call framing. The attack union includes size/order, timing, combined features and learned flush removal. Seven shared learners, including a 1D-CNN and forest-leaf neighbors, also permit a matched comparison with offline website-fingerprinting (WF) adaptations.

Timing reveals distinguishable cover. For standard-prompt serial shadow, adding timing/flush attacks increases cancellation BA from 50.0% to 61.4% (paired increase 11.4 points, 95% interval [3.7,19.6]); return changes from 50.7% to 53.4%. Flush-phase detection reaches 90.4/92.3% for serial/parallel schedules over 15,278 records from 276 test episodes. A separate cover-versus-real request classifier reaches 82.5/96.1% on 7,639 requests.

We then compose the detector with intent inference: a training-fitted detector strips the predicted suffix before classification. Table 17 shows the recovered category signal, with a strongest return point of 60.7%. The full attack union retains both stripped and unstripped traces. This separates the ability to recognize padding phases from the ability to recover a task's operation.

Match cost to the protected boundary. Measured real-chain p95 latency is 26.2/20.8 ms for serial/two-in-flight shadow, versus 56.4/112.3 ms for 2/5 ms tunnels. Sessions preserve real responses and final live state; fixed tunnel sizes/counts still expose host-scheduled direction/order. Table 16 keeps measured RPC times separate from offline schedule transformations.

Under the common seven attacks, random-donor Walkie-Talkie (WT) reaches 50.0/56.7% BA at 2.04× bytes, versus serial shadow's 50.0/52.9% at 15.11×. WT also matches or improves both BA points, bytes and calls over parallel shadow; under rules, its 51.4/56.3% at 2.23× improves those four point metrics over both shadow schedules. These evaluated point trade-offs motivate choosing protection at the narrowest required boundary.

The WF adapters use the same saved traffic and training customers. WT samples donor traces without symmetric site pairing; WTF-PAD samples training inter-arrival histograms; Tamaraw and FRONT supply fixed-rate and front-loaded controls. Input-derived fragments preserve sizes, directions and order, enabling a controlled metadata comparison. Their simulated spans and measured RPC latency remain separate cost objects.

TABLE 13. EXP 3 FROZEN: FROZEN-CALIBRATION OPERATION-CATEGORY DIAGNOSTICS ON REAL-AGENT TRAJECTORIES. TESTING COVERS 46 TASKS FROM 24 HELD-OUT CUSTOMERS, TWO PROMPTS AND THREE REPEATS (276 EPISODES). NETWORK BA IS THE MAXIMUM OVER THE FIXED ATTACK SUITE, WITH CUSTOMER-CLUSTER 95% INTERVALS.

Condition	Cancel BA [95% CI]	Return BA [95% CI]	Reward (%)	Bytes×	Calls×
Undefended	97.2 [90.0, 100.0]	94.0 [85.8, 100.0]	79.7	1.00	1.00
Rules prompt	97.2 [90.0, 100.0]	93.5 [85.0, 99.5]	76.1	1.00	1.00
Padding only	66.1 [50.0, 83.9]	55.5 [50.0, 73.0]	79.7	4.02	1.00
Shape only	60.8 [55.4, 71.3]	51.9 [50.5, 57.8]	79.7	4.16	3.83
Joint program	50.0 [50.0, 60.6]	50.0 [50.0, 65.5]	79.7	15.39	3.83
Rules + joint	50.0 [50.0, 65.1]	60.6 [50.0, 73.3]	76.1	15.72	3.88

Paired joint-minus-undefended BA differences (percentage points): standard-prompt cancellation -47.2 [-50.0, -33.3], return -44.0 [-48.9, -27.8]. These post hoc contrasts use the same customer draws and reselect the attack-suite maximum in each condition. Joint = online shadow cover plus constant framing; padding/shape ablations use standard-prompt trajectories. Ratios use undefended transport within each prompt. Shadow isolation preserves the logical planner transcript; the four program conditions replay actual LLM trajectories over local TLS 1.2. Reward is measured on original trajectories and inherited by program transforms; see the utility scope in §9.3.

TABLE 14. EXP 3: EXPLORATORY CROSS-FIT COMPARISON UNDER THE EXPANDED 11-ATTACKER UNION: ALL 690 EPISODES AND 53 CUSTOMERS. MAXIMUM NETWORK BA WITH CUSTOMER-BOOTSTRAP 95% INTERVALS. STANDARD PROMPT UNLESS MARKED RULES; BYTE/CALL RATIOS RELATIVE TO UNDEFENDED TRANSPORT.

Configuration	Cancel BA (%)	Return BA (%)	B×/C×
Undefended	91.7 [82.1, 100.0]	85.5 [76.7, 93.4]	1.0/1.0
Joint constant	60.5 [52.9, 72.4]	55.5 [51.8, 64.2]	19.8/5.0
Joint buckets	57.1 [50.0, 71.5]	52.7 [50.0, 61.7]	12.4/5.0
Rules + buckets	56.9 [50.2, 71.2]	52.7 [50.0, 65.0]	12.4/5.0

Joint constant is Exp 3 frozen's shadow-count plus constant-framing design, refit within each fold. Joint buckets changes framing to 1024-byte buckets. Every bootstrap draw reselects the complete maximum. Intervals condition on fitted predictions.

TABLE 15. EXP 3 ORACLE LOCALIZATION UNDER THE 11-ATTACK UNION. STANDARD PROMPT: 345 CROSS-FITTED EPISODES, 53 CUSTOMERS. MAXIMUM NETWORK BA (%) WITH CUSTOMER-BOOTSTRAP INTERVALS.

Target	Full trace	Writes removed	Writes only
Cancel	91.7 [82.1, 100.0]	62.8 [54.6, 75.3]	91.7 [82.1, 100.0]
Return	85.5 [76.7, 93.4]	59.9 [52.2, 71.8]	84.9 [78.8, 94.0]

Finding. A length-only verdict misses schedule leakage, and expensive shadow execution has no measured network-efficiency advantage over all transport alternatives. For a passive ingress-link observer, lightweight shaping is a relevant candidate; for a restricted dispatch observer, execution isolation addresses an additional state boundary. This is the audit's placement-and-cost distinction demonstrated on one workload.

Synthesis. The demonstrations operationalize the audit: match the emitter before borrowing a byte cost (Exp 1), verify executed state before borrowing a cover strategy (Exp 2), bind targets and calibration to a shared cohort (Exp 3), and price timing protection against matched transport alternatives (Exp 4). They show how source-bound comparisons change a concrete deployment decision.

9. Implications, Ethics and Limits

9.1. Deployment paths and reporting obligations

A passive agent–tool observer can be an access-network operator, an authorized VPC flow monitor, or a relay on an inter-organization path. Each position defines a specific observable projection. Table 18 states the corresponding visibility and trusted endpoint. Encrypted ClientHello hides the inner server name when supported, but destination IPs and downstream connections can still identify tools [73]. Our measurements use a single ingress endpoint.

Shadow state has three possible implementations. **(1) Authorized reads and verified no-ops** use the backend with discarded responses; idempotence alone does not preserve state. **(2) Provider-supported isolation** includes Kubernetes server dry-run, separate Stripe sandboxes and transactional rollback [74], [75], [76]. These have different semantics: for example, sequence changes can survive rollback [77]. **(3) A trusted tool gateway** authenticates cover markers and routes shadow writes to an isolated executor. Our local measurements implement the gateway path; the provider-specific paths identify deployments for follow-up measurement.

The trusted gateway/executor can identify cover. Protection ends at exposed markers, routing branches or commit logs; it does not protect against that principal. On the ingress link, the inexpensive WF adaptations are relevant competitors. At a restricted dispatch view, hiding call profiles additionally requires marker/routing isolation. The choice therefore follows the required observation boundary. Cover also consumes quotas and shared capacity; admission budgets, backpressure and separate credentials must prevent overload or accidental live writes.

9.2. Ethics and artifact scope

Executions use cloned public benchmark databases and loopback sockets; model requests contain public tasks and simulated interactions. No production account is modified or third-party service probed for a vulnerability. Stakeholders

TABLE 16. TIMING-AWARE EXP 4 COMPARISON: STANDARD PROMPT, 138 HELD-OUT EPISODES, 24 CUSTOMERS (5 CANCEL-POSITIVE, 7 RETURN-POSITIVE). EACH BA PAIR IS CANCEL/RETURN (%). THE COMMON SEVEN MODELS PERMIT A MATCHED-SUITE COMPARISON; THE FULL UNION ALSO RETAINS ALL EARLIER ATTACKS.

View	Condition	Common-seven BA	Full-union BA	Upper BA	Bytes×	Calls×	RPC p95 (ms)
M	Undefended	84.3/76.9	91.7/86.4	100.0/97.2	1.00	1.00	15.4
M	Per-call constant	53.2/59.1	68.6/62.7	92.8/79.5	3.92	1.00	16.6
M	Serial shadow	50.0/52.9	61.4/53.4	71.4/65.8	15.11	3.83	26.2
M	Two-in-flight shadow	50.0/57.3	51.7/57.7	69.1/69.7	15.11	3.83	20.8
M	Full tunnel, 2 ms	59.9/55.2	62.5/55.2	70.2/62.0	125.71	1.00	56.4
M	Full tunnel, 5 ms	54.6/53.7	54.6/57.3	72.9/64.2	125.71	1.00	112.3
S	Record buckets	53.1/69.8	53.1/69.8	78.5/86.9	2.25	1.00	–
S	Tamaraw 40/12 ms	55.3/55.3	55.3/55.3	68.7/64.4	12.15	1.00	–
S	Tamaraw 4/1.2 ms	57.8/51.8	57.8/51.8	71.8/61.8	11.99	1.00	–
S	FRONT	59.0/69.3	59.0/69.3	69.9/81.8	2.51	1.00	–
S	WTF-PAD adaptation	51.9/62.7	51.9/62.7	64.1/79.6	1.88	1.00	–
S	WT random-donor	50.0/56.7	50.0/56.7	67.0/67.8	2.04	1.00	–
S	FRONT, short calibration	52.4/56.0	52.4/56.0	61.3/71.1	2.33	1.00	–

M: measured local tool RPC on saved actions; S: offline traffic transformation. RPC p95 ends at the last real response, excluding LLM/WAN. S-row simulated spans are not RPC latency. Upper BA is the full-union 95% bound. M unions contain 17–22 attacks; S unions contain the common seven. Ratios use prompt-specific undefended captures. The utility scope is defined in §9.3.

TABLE 17. EXP 4 INTENT BA AFTER LEARNED FLUSH REMOVAL (95% CUSTOMER-CLUSTER INTERVAL). EACH ROW: 138 TEST EPISODES, 24 CUSTOMERS. MAXIMUM OF THREE CLASSIFIERS PLUS MAJORITY; THE COMPLETE ATTACK UNION ADDITIONALLY KEEPS UNSTRIPPED OBSERVATIONS.

Prompt	Schedule	Cancel	Return
Standard	Serial	57.8 [50.0, 70.7]	53.4 [50.0, 64.6]
Standard	Parallel	50.0 [50.0, 64.0]	50.0 [50.0, 56.7]
Rules	Serial	50.1 [50.0, 55.6]	60.7 [50.8, 70.9]
Rules	Parallel	50.8 [50.0, 61.7]	54.0 [50.0, 67.0]

TABLE 18. DEPLOYMENT BOUNDARY: THE RELEVANT PASSIVE OBSERVER AND WHAT THE MEASURED MECHANISM CAN PROTECT.

Setting	Visible information	Applicability / trusted point
Same-host sidecar	Host trace/process/IPC metadata, if permitted	Link padding alone does not hide host instrumentation; isolate the monitor's access.
Same-VPC gateway	Ingress record sizes, times and addresses	Tunnel or shadow shaping can protect this link; gateway/markers remain trusted.
Remote SaaS/MCP	IP, unprotected SNI, per-endpoint flows and times	Shared ingress needed for tool-identity cover; ECH alone does not hide destination IP.
Cross-organization	Each hop's routing; provider calls/commit log	No guarantee across all hops. A trusted isolated executor is an explicit added assumption.

include paying tenants, shared-capacity providers and people affected by state changes. Other-account cancellation is a simulator control. Released inference code can support surveillance; examples use public/synthetic identities and controlled endpoints. Bot-detection evasion has a different authorization boundary [12], [6].

AI assisted candidate-field extraction, implementation, source reconciliation and prose revision. Authors retain responsibility for the source-linked comparison and its interpretation. Authors' source verification is specified in §2.3. Saved-output replay, model-generated trajectories and measured local TLS are distinguished throughout. The artifact archives source locators, decisions, observations, calibration and executable checks. An anonymous submission snapshot must omit identity metadata and development history; the separate submission checklist records the AI-use disclosure and author review required at registration.

9.3. Threats to validity and scope of evidence

Source synthesis. The purposive corpus supports comparisons within the assembled evidence base; seed-set query coverage is not field-wide recall. Of 77 cited entries, 49 are arXiv-only, 21 venue-labelled (five also cite preprints), and seven are books, standards or vendor documents. These labels identify source form. Initial AI extraction and author adjudication are dependent stages, so their archived agreement measures process consistency rather than independent human reliability. Constructed cases similarly check rule execution. Source locators and applicability-sensitive denominators support direct scrutiny; independent blind-recoding templates are available for extension.

Statistical scope. The original BA/AUROC screen is an author-chosen engineering threshold, not a security definition. No reported joint design confirms that two-target absolute criterion; the measured paired reductions establish improvements against specified fitted suites. Confidence intervals reselect the suite maximum but condition on fitted predictions and omit retraining uncertainty. The frozen split has only 5/7 positive customers and the full cohort 14/18; repeats add execution coverage, not independent customers. Reference-program calibration uses the known population. Real-agent calibration is frozen before held-out collection, while development had access to the full task set; the

cross-fit, timing follow-ups and sensitivity analyses are exploratory known-cohort evidence.

Targets and deployment utility. The operation-category targets are coupled to intended actions. Their localization establishes operation signatures, not latent customer-attribute recovery. A small same-operation product diagnostic is inconclusive (Appendix C). Our utility audit distinguishes measured original agent reward, checked defended logical state/response conformance, and unmeasured live defended-planner utility. Local RPC timing excludes LLM generation, WAN transit and timing-dependent replanning. WF adapters support matched metadata comparisons; original-protocol equivalence and equal-utility deployment comparisons require further measurements.

External validity. Sol and Luna are mutable aliases from one provider in one retail environment, each also used in a separate user-simulator context. The additional Luna study has 29 training episodes per prompt. Fresh independent customers, fixed open weights, cross-vendor and cross-environment tests, provider-backend measurements and live replanning are the next validation axes. The present synthesis supplies the source bindings and comparison obligations needed to design those tests.

10. Conclusion

This SoK makes the evaluated result record the unit of defense comparison. Its 82-family map, 39 source-bound records and nine-item audit expose transfers hidden by paper-level grouping: protection lost at a new emitter, read cover moved across a state-changing boundary, and utility or cost borrowed from another mechanism. Four demonstrations show why those distinctions change deployment choices. They separate compliance from operation privacy, expose scheduling leakage, and identify inexpensive transport alternatives to costly shadow execution on a common workload. The practical contribution is an inspectable path from published evidence to a justified comparison: identify the receiver, preserve the intended state, and bind inference, utility and cost to the same operating point.

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Appendix A.
Audit Trail and Logical Equivalence

Evidence organization. The package separates the 82-family source map, 39-result ledger, 180-cell audit, transport tests, dispatch tests and real-agent studies. Each source record has a version and section/table/figure/paragraph locator. Authors reviewed all family labels and verified the result records and audit cells as described in §2.3. Initial AI-assisted extraction and subsequent author adjudication remain distinct archived columns; the supplementary process statistics and blind-recoding templates support follow-up audit. The supplement supplies detailed algorithms, full result grids and archived coding decisions; the main paper contains the comparison rules, complete family matrix and record index needed to follow the synthesis.

Logical equivalence. For deterministic tools, a real action gives $(s_{t+1}, o_t) = T(s_t, a_t)$. The defended dispatcher forwards that action unchanged, runs cover on a disjoint state s_t^{sh} and suppresses cover responses. With equal initial states and realized real actions, induction gives

$$s_t^D = s_t, \qquad \pi_{\mathrm{plan}}(D(H_t)) = H_t.$$

Here H_t is the logical message/action/response history, excluding wall-clock time. Payload-preserving framing and terminal shadow flush maintain the invariant, provided timeouts, shared-state side effects and timing-dependent tool behavior are absent. Checks compare every real-call projection and scoring field across all 690 trajectories and four program conditions. These establish implementation conformance to isolation. The utility boundary in §9.3 explains the further live-planner measurements.

The reproducibility index resolves each archived experiment in Appendix C to protocol, inputs, predictions and checksums. Saved-output verification reproduces the reported metrics without new model calls, while measured captures and analytic reconstructions retain their separate provenance.

Appendix B.
Result-record index

Table 19 indexes all 33 retained baseline and six expansion records. Separate records identify changes in observation, intervention, target, execution population or cost object within a family. Shortened source locators below support direct lookup; the machine-readable ledger retains exact versions, hashes and full locators.

Appendix C.
Experiment artifact index

Table 20 maps the four main experiments to stable artifact keys and detailed supplementary methods. Reanalysis, prompts and repetitions retain their original customer/task membership.

The reproducibility index resolves these keys to files, protocols, saved predictions and executable verification commands. The source versions and calibration chronology remain available for every analysis.

Extensions and sensitivity. The same-provider Luna extension contains 150 formal runs on 75 existing retail tasks, two prompts and one repeat. Standard-prompt write-only dispatch BA is 87.5/66.6%, versus 50.0/57.0% for get/find/list reads; rewards are 32/75 and 33/75, with seven protocol-error episodes retained. This tests the operation diagnostic under another model alias.

A customer-composition check on fixed cross-fit predictions changes standard-prompt joint-constant BA from 60.5/55.5% to 59.0/53.2% under equal-customer-within-class weighting. Rules-arm return changes from 59.4% to 60.4%. Across 192 leave-one-positive-customer checks, standard-prompt cancellation ranges 57.3–62.0%; these are influence ranges, not confidence intervals.

The same-operation attribute check fixes single-item returns and a pre-fit digital-device grouping. Per prompt, 20 customer–order–item instances from 15 customers use the original customer folds and fixed LR/RF/GB learners. Maximum OOF BA is 30.3% [16.7,52.7], versus 50% for a constant predictor; a single fixed shuffled target gives 65.2%. Both prompts have identical features. Below-chance predictions can reflect reversed associations, and one shuffle is not a permutation distribution. This diagnostic is inconclusive for non-action attributes; its full predictions remain in the supplement.

TABLE 19. COMPLETE INDEX OF THE 39 RESULT RECORDS (26 FAMILIES). EACH ROW RETAINS ITS ARTIFACT ID, FAMILY CITATION, OBSERVATION/EXECUTION BOUNDARY AND SOURCE-LOCAL LOCATOR. THE PANELS FORM ONE INVENTORY, NOT INDEPENDENT DATASETS. T=TABLE, F=FIGURE, APP.=APPENDIX; LOCATORS REFER TO THE CITED SOURCE.

Record ID / family	Boundary, distinction; source locator
UI-delay-offline [5]	Offline delay; no live task outcome. *6.1 F5; A.5*
WRA-decoy-execution [6]	Executed decoys; usefulness proxy only. *8.1-8.3 T8-11*
NET-upstream-local [36]	Protected deployment/local serving; topology scoped. *IV-C-E TIII-IV*
NET-dummy-padding-simulation [36]	Synthetic insert/pad; no live execution. *VII TIX*
EARLY-K-calibration-oracle [3]	K-specific fit then oracle; not live KV trial. *V-A TVI; IV-A*
EARLY-semantic-deidentification [3]	GPTCache prototype cost; not KV-granularity cost. *V-B*
WHISPER-randompadding-capture [39]	New padded captures; rounded medians. *6.1 T3-4*
WHISPER-batching-replay [39]	Offline batching; model-dependent result. *6.2 F7*
WHISPER-injection-replay [39]	Synthetic packets; no live zero-delay claim. *6.3 T5*
SPLICE-CTBF-signal [46]	Prototype background load; tested signal only. *VII-C TII*
SPLICE-QCP-cost [46]	QCP component cost; excludes combined CTBF. *VII-C TII*
TIME-length-instruction [47]	Length instruction; unequal outputs remain. *7.2*
TIME-padding-estimate [47]	Analytic plaintext cost; not network bytes. *7.3*
SPEC-padding-REST [38]	Padding/aggregation; payload cost, placement unverified. *6.1 F6-7 T2*
DANGOL-capture [14]	Capture transform; distinct from live population. *9.8-9.9 T6*
DANGOL-livebinding [14]	Live loopback binding; do not join capture accuracy. *9.7, OHTTP paragraph after T 5*
SIG-common-guardrail [55]	Common instruction; not a causal guardrail effect. *4.1-4.2; App.C*
SAFE-post-trigger [15]	Post-trigger probes; detector cost not service TTFT. *7.2.4 F12; 7.3-7.4*
AHB-online-executed [12]	Executed UI edits; task/state changes possible. *5.1.2 T1; E.1-E.3*
AHB-offline [12]	Offline touch edits; no online task damage claim. *T1 offline rows; E.1-E.3*
KVGOV-partialflush-sim [33]	Deterministic simulation; not live protected recovery. *5.1-5.3*
KVGOV-hardware-gap [33]	Hardware gap calibration; not protected ASR. *5.7-5.8*
KVGOV-HMAC-cost [33]	HMAC microcost; not cache reuse/task loss. *5.9*
TRAFFIC-prompt-reformulation [51]	Collected prompt rewrites; task utility unmeasured. *V-C1 F9a*
TRAFFIC-temperature [51]	Collected decoding changes; no monotone benefit. *V-C1 F9a*
CARLINI-fixed-rate [35]	Rate/replayed cost; per-token, not request delay. *VII F11-12*
GAO-zero-copy-off [20]	Loader component costs; not end-to-end TTFT. *7.1*
TDXRAY-oblivious-map [27]	Lookup microbenchmark; not full inference cost. *6.1 F8 T3*
AGENTTELL-fixed-instruction [62]	Live local browser tasks; loaded/cold context contrast. *4.3.1; 5 Finding1*
FP-CLOUDFLARE [63]	Live access control; not user confidentiality. *6.4 T5*
BAR-co-running-noise [21]	MIG noise test; not an LLM defense trial. *7.2-7.3 T8*
RHYTHM-VPN [34]	Live VPN shift; baseline-trained attacker only. *VI-B; VII*
DEB-filter-origin [42]	Filter creates attack view; not mitigation of it. *Threatmodel; 4.2*
CACHEPRUNE-selective [48]	Selective cache sharing; direct token recovery/quality. *3.1; 5.1–5.6; T1; App.A*
CACHEPRUNE-context [48]	Contextual guessing; distinct from direct recovery. *3.1; 5.1–5.6; T1; App.A*
OTRO-unpadded [16]	ORAM TTFT/memory; access-count padding excluded. *VII-C–E; TIII*
OTRO-epochpad [16]	Epoch-padding estimate; not measured padded TTFT. *VII-C–E; TIII*
SPARLEAK-replacement [25]	Sparse replacement/quality; attack cost is separate. *3.1; 4–7; T4*
SHAPE-padding-estimate [26]	Constant-work proposal/estimate; no executed pair. *IX-B1*

TABLE 20. FOUR MAIN-TEXT MODULES AND THEIR HISTORICAL ARTIFACT KEYS. THE RIGHT COLUMN SUPPLIES OPTIONAL IMPLEMENTATION AND FULL-GRID DETAIL.

Main module	Artifact keys	Observation/execution boundary and comparison	Detailed source
Exp. 1: transport emission	E1, E3, E5	Buffered two-link re-emission and token-stream emitters; upstream framing, final-link framing and matched attacker refitting are distinct comparisons.	S9/S11; S12
Exp. 2: execution and link	E2, E6, E7	Prescribed stateful plans and randomized dispatch cover, then agent–tool TLS composition; dispatch-visible calls and encrypted-link records are separate views.	S9/S11; S12
Exp. 3: real trajectories and localization	E9, E10, E12	Saved autonomous Sol trajectories; original pre-test calibration, exploratory customer cross-fitting and oracle operation-category localization retain their different fit/test boundaries.	S14; S15/S21
Exp. 4: time and WF	E13, E14	Local live tool-loop timing/full tunnels, then stronger learners and specified WF adaptations; measured clocks and simulations are distinguished, with matched suites for joint comparisons.	S19; S20
Extensions	E8	Stand-in-planner pipeline check.	S13
	E11	Retrospective training-fold operating-point selection with the historical suite.	S15
	E15	Additional same-provider Luna cohort.	S22
	E16	Same-operation semantic-attribute diagnostic on reused Sol data.	S23.8